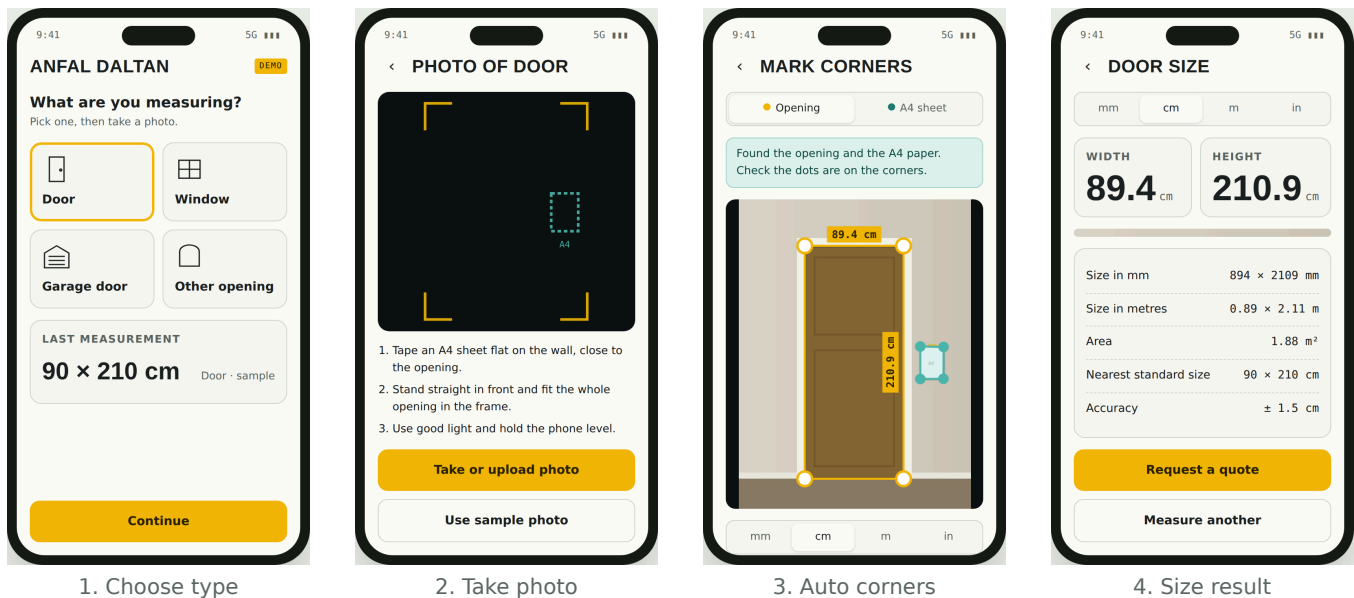


Anfal Daltan Door & Window Measurement App

A mobile app for customers: take one photo of a door, window or opening, and the app returns the width and height in mm, cm, m or inches. The measurement is sent to our team for a quote.

Version 1.0 · September 2026 · Prepared by Muhammad Khan



Screens from the working clickable prototype.

1. What the app does

1 Choose Door, window, garage door or other opening.	2 Capture Photo straight on, with an A4 sheet on the wall, or AR scan on supported phones.	3 Detect AI finds the opening corners and the reference sheet. Customer can fine-tune with arrows.	4 Measure Width, height, area, nearest standard size in mm / cm / m / in.	5 Quote Photo + size sent to the sales team; customer gets a quote.
--	--	--	---	---

Prototype status (done)

- Clickable phone prototype with all 4 screens.
- Measurement from a photo using an A4 sheet as the scale.
- Automatic detection of the door frame and the A4 sheet (rule-based computer vision).
- Dot fine-tuning with magnifier and arrow buttons.
- Units: mm, cm, m, inches; area in m²; nearest standard size.
- Warnings for wrong photos (sheet missing, dots on the wrong object).

The production app replaces the rule-based detection with trained AI models and adds AR measuring, accounts, quotes and a sales dashboard.

2. Measuring methods

The app offers the best method the customer's phone supports, and falls back to the next one.

Method	How it works	Phones	Expected accuracy*
A. AR + LiDAR	Customer taps the corners in the live camera view; the depth sensor measures real distance in 3D. No reference sheet needed.	iPhone Pro / iPad Pro	± 0.5–1 cm
B. AR (camera only)	ARCore / ARKit tracks the wall plane from camera movement; corners are placed on that plane.	Most Android and iPhones from the last ~6 years	± 1–3 cm
C. Photo + A4 sheet	One photo with an A4 sheet (210 × 297 mm) on the wall. AI finds corners; the sheet gives the scale; perspective is corrected.	All phones	± 1–2 cm
D. Photo + known size	Customer draws a line on something of known length (tile, tape measure) and types its length.	All phones	± 2–4 cm

* Typical values for a photo taken straight on in good light. To be confirmed by testing during development.

3. AI and computer-vision models needed

Model	Job in the app	Suggested technology	Priority
1. Opening detection	Finds the door / window / opening in the photo and outlines it (segmentation mask).	YOLO11-seg or YOLOv8-seg (Ultralytics), trained on door & window photos	Must
2. Corner (keypoint) model	Places the 4 corners of the opening precisely, including the inside edge of the frame.	YOLO11-pose with 4 keypoints, plus edge refinement with OpenCV	Must
3. Reference object detection	Finds the A4 sheet (or a bank card) and its 4 corners to get the scale.	OpenCV contour detection; small YOLO model as backup	Must
4. Perspective correction	Straightens photos taken at an angle so the size stays correct.	Homography with OpenCV (math, no training)	Must
5. Photo quality check	Before measuring: checks blur, darkness, phone tilt, wide-angle lens, whole opening in frame.	OpenCV (blur score) + phone gyroscope + camera lens info	Must
6. Type classifier	Recognises door, window, garage door, arch; picks the right standard sizes.	Class output of model 1, or MobileNetV3 classifier	Should
7. AR measuring	3D measuring with the live camera (methods A and B).	Apple ARKit (iOS), Google ARCore with Depth API (Android)	Should
8. Depth estimation	Extra check of distance for phones without AR support.	Depth Anything V2 (metric) – lower accuracy, used only as a cross-check	Optional
9. AI assistant	Writes the quote text, answers customer questions, reads notes (e.g. "needs frame too").	Large language model API (e.g. Claude)	Optional

Running the models

- **On the phone (preferred):** models 1–6 exported to TensorFlow Lite (Android) and Core ML (iOS). Fast, works offline, photos stay private.
- **On the server (backup):** same models with ONNX Runtime or PyTorch for older phones, and to re-check results before a quote.

Training data

Item	Details
Photos needed	About 3,000–5,000 photos of doors, windows and openings: different rooms, lights, frames, open/closed doors, with and without A4 sheet.
Sources	Public datasets (e.g. door/window sets on Roboflow Universe), our own site photos, test photos from staff.

Labelling	Outline + 4 corners per opening. Tools: CVAT, Roboflow or Label Studio.
Ground truth	For 200+ test photos, measure the real size with a tape measure to test accuracy.
Training	Cloud GPU (e.g. 1× NVIDIA T4/A10). Re-train every few months with new customer photos (with permission).

4. App components

Component	What it includes	Suggested technology
Mobile app	iOS and Android app from one code base: camera, AR, on-device AI, measuring screens, history, quote requests, Arabic + English.	Flutter (or React Native)
Camera & AR	Guided capture (level indicator, frame guide), AR corner tapping.	camera plugin, ARKit / ARCore plugins
Backend API	Customer accounts, measurements, photos, quotes, notifications.	Python FastAPI or Node.js
Database & storage	Customers, measurements, quotes; photo storage.	PostgreSQL; S3-compatible storage
AI service	Server-side model checks and re-training pipeline.	Python, PyTorch, Ultralytics, OpenCV, ONNX
Sales dashboard (web)	Team sees new requests, photo with measured outline, edits size, sends price.	React / Next.js web app
Notifications	Quote ready, appointment reminders.	Firebase Cloud Messaging, WhatsApp Business API, email / SMS
Login	Phone number with OTP.	Firebase Auth or similar
Hosting	Servers, database, backups, monitoring.	AWS / Google Cloud / Azure (or Render for early stage)
Analytics	Usage, errors, measurement success rate.	Firebase Analytics, Crashlytics / Sentry

Screens

- Welcome and login (phone + OTP) · Language (Arabic / English)
- Choose type · Capture guide · Camera / AR measure · Check corners
- Result (units, area, standard size) · Add notes and more openings · Request quote
- My measurements (history) · Quote details · Profile and settings
- Dashboard (web): requests list · request detail with photo · price editor · customers

5. Development plan

Phase	Scope	Estimate
1. MVP	Mobile app with photo + A4 method (C), auto corners with fine-tuning, units, quote request, simple sales dashboard, Arabic + English.	8-10 weeks
2. AI models	Collect and label photos, train models 1-3, on-device deployment, perspective correction, photo quality check.	6-8 weeks (can run alongside phase 1)
3. AR measuring	Methods A and B with ARKit / ARCore, automatic choice of the best method per phone.	4-6 weeks
4. Launch	Accuracy testing on 200+ real openings, bug fixes, App Store and Google Play release.	2-3 weeks
5. After launch	Re-train models with new photos, add types (arches, sliding doors), AI assistant, price calculator.	Ongoing

Estimates assume the team below and may change after requirements are confirmed.

Team

- 1 mobile developer (Flutter, camera, AR)
- 1 backend / web developer (API, dashboard)
- 1 computer-vision / ML engineer (models, training, accuracy)
- 1 UI/UX designer (part time) · 1 tester (part time)

6. Accuracy and limits

Why a reference is needed. A single photo cannot show real size: a small door close up and a big door far away look the same. The app therefore needs either a known object in the photo (A4 sheet), or depth information from AR / LiDAR.

- Photos must be taken straight on, with the normal 1× lens (not 0.5× wide-angle).
- The A4 sheet must be flat on the same wall as the opening.
- For manufacturing, the app size is used for the quote; final sizes are confirmed on site before production.

7. Next steps

- Confirm which size the client needs: door leaf, opening inside the frame, or outside of the frame.
- Confirm app name, branding and languages.
- Collect 50-100 real photos with tape-measured sizes to test the prototype.

- Approve phase 1 scope and start development.